

6 Conic Sections

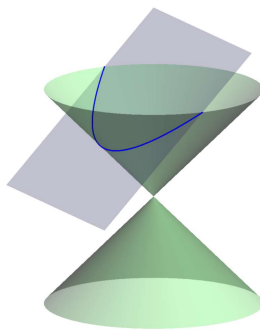
We consider the traditional conic sections and relate their basic properties to modern equations.

Definition 6.1. A (non-degenerate) conic section is the intersection of a right circular cone²² with a plane not containing the vertex. Depending on the orientation of the intersecting plane, we obtain one of the three standard types.

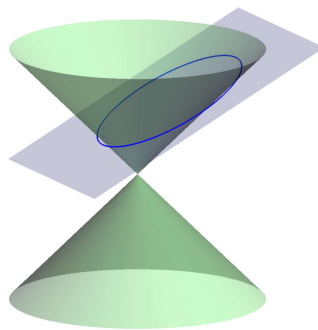
Parabola The plane intersects in a single curve either on the upper or lower half of the cone. Equivalently, the plane is parallel to one of the lines defining the cone.

Ellipse The intersection is a single closed curve.

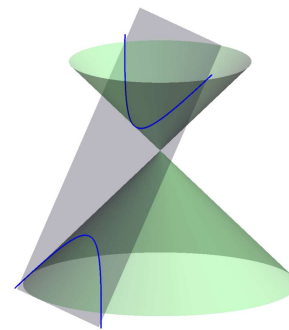
Hyperbola The intersection comprises two curves, one each on the upper and lower halves of the cone.



Parabola



Ellipse



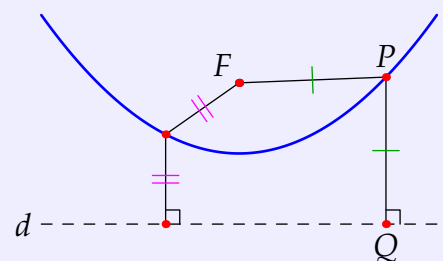
Hyperbola

6.1 The Focus & Directrix of a Parabola

The above conic section definitions are hard to work with because they are three dimensional. We instead make alternative definitions as curves in the plane.

Definition 6.2. Let d be a line and F a point not on that line. The *parabola* (in the plane spanned by d and P) with *directrix* d and *focus* F is the set of points P equidistant from both. As pictured, the parabola is the set of points for which

$$|FP| = |PQ|$$



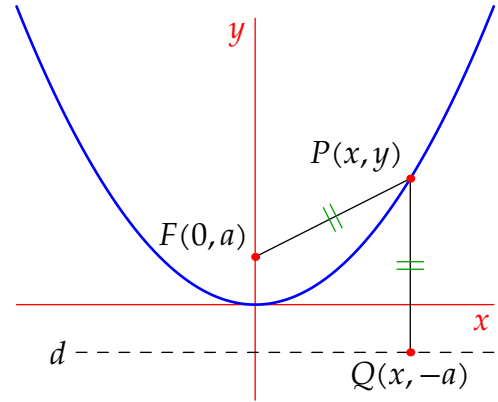
The definition is strictly *synthetic* since equidistant really means that the line segments $\overline{FP}$ and $\overline{PQ}$ are *congruent*. To interpret this *analytically* we choose axes and co-ordinates.

²²A *cone* is a surface obtained by joining all points on a curve with a fixed point, the *vertex*. A right circular cone is when this curve may be chosen to be a *circle* such that the line from its center to the vertex (the *axis*) is perpendicular (*right angles*) to the circle.

Without loss of generality, choose the origin to lie on the parabola halfway between the focus and directrix, with the focus on the positive y -axis. Otherwise said, for some $a > 0$, we have $F = (0, a)$ and the directrix is $y = -a$.

Now we follow the definition: if $P = (x, y)$ is a point on the parabola, then $Q = (x, -a)$, whence

$$\begin{aligned} |FP| = |PQ| &\iff |FP|^2 = |PQ|^2 \\ &\iff x^2 + (y - a)^2 = (y + a)^2 \\ &\iff x^2 + y^2 - 2ay + a^2 = y^2 + 2ay + a^2 \end{aligned}$$



This simplifies to recover a familiar equation:

Theorem 6.3 (Canonical form of a parabola). *Given a parabola, there exist rectangular co-ordinates/axes with respect to which the parabola has equation*

$$x^2 = 4ay$$

where $a > 0$.

Of course, parabolas are often not always presented in canonical co-ordinates.

Example 6.4. Find the focus and directrix of the parabola with equation $y = x^2 - 6x + 7$. Start by completing the square!

$$y = x^2 - 6x + 7 = (x - 3)^2 - 2$$

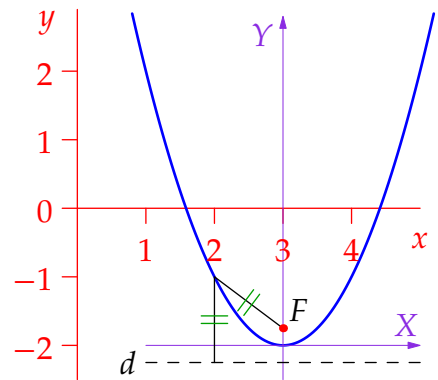
By defining new co-ordinates $X = x - 3$ and $Y = y + 2$, we obtain the canonical form

$$X^2 = Y \quad \text{where} \quad a = \frac{1}{4}$$

The focus has co-ordinates $(X, Y) = (0, \frac{1}{4})$ and the directrix is $Y = -\frac{1}{4}$. Translating back to our original co-ordinates (x, y) , we see that the focus and directrix are

$$F = \left(0 + 3, \frac{1}{4} - 2\right) = \left(3, -\frac{7}{4}\right)$$

$$d : y = -\frac{1}{4} - 2 = -\frac{9}{4}$$



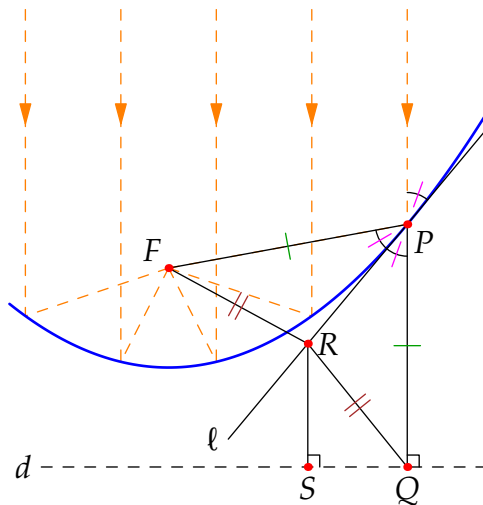
Canonical co-ordinates can also be rotated relative to given axes, but this would take us too far afield. You will likely return to such ideas in an upper-division linear algebra course.

Why call it the *Focus*?

Imagine a *parabolic mirror*: a reflective surface obtained by rotating a parabola around its canonical y -axis. The rays from a light source at 'vertical infinity' (e.g., the sun) would arrive parallel to the y -axis. The parabolic mirror will *focus* these rays onto the focus of the original parabola! This is the reason satellite dishes are parabolic with the receiver placed at the focus.

Convincing ourselves of this requires only a little triangle geometry. Let P lie on the parabola, Q be the nearest point on the directrix, and bisect angle $\angle FPQ$ with the line ℓ . Let $R \neq P$ be any other point on ℓ and drop the perpendicular to d at S .

1. Side-Angle-Side for $\triangle FPR \cong \triangle QPR$ tells us that $|FR| = |RQ|$.
2. Since $\overline{RQ}$ is the hypotenuse of a right-triangle, we see that $|RS| < |RQ| = |FR|$.
3. All points $R \neq P$ on ℓ therefore *lie on the same side of the parabola* (closer to d than F), whence ℓ is *tangent* to the parabola.
4. A vertical ray hitting the parabola at P is reflected so that its angle of incidence equals its angle of reflection. Since ℓ is tangent, this angle is $\angle QPR \cong \angle FPR$. The ray is therefore reflected to $\overline{PF}$ and passes through F .



Exercises 6.1. Key concepts: *Slicing a cone with a plane, Focus and directrix definition of a parabola, Canonical form by completing the square, Meaning of focus*

- Put each parabola in canonical form, and thus find its focus and the equation of its directrix. Sketch each parabola.
 - $x^2 + 2x - y = 0$
 - $4y = x^2 - 6x + 1$
 - $4y^2 - 3x + 4y + 1 = 0$
- A common design for a flashlight places a lightbulb at the focus of a parabolic mirror. Explain why. What would happen to the light beam if the bulb were placed just a little further than the focus from the mirror?
- A right-circular cone centered on the z -axis has equation $z^2 = x^2 + y^2$ (horizontal cross-sections are circles).

Intersect this cone with the plane with equation $z = y + k$ for some constant k . Show that the resulting curve is indeed a parabola. Determine its focus and directrix, and the value of a in its canonical form.

(Hint: You'll need to consider a factor of $\sqrt{2}$ with y to compensate for the slope of the intersecting plane—why?)

6.2 Eccentricity: Ellipses and Hyperbolæ

The words *hyperbola* and *ellipse* have connotations of “more than” and “less than.” But *more* or *less* than what?²³ The answer lies in a simple modification of Definition 6.2.

Definition 6.5 (Conics via Eccentricity). Let d be a line and F a point not on that line. Let $e > 0$ be a constant. The *conic* with focus F , *directrix* d and *eccentricity* e is the set of points P for which

$$|FP| = e |PQ|$$

where $Q \in d$ makes a perpendicular $\overline{PQ} \perp d$. The value of e determines the type of conic:

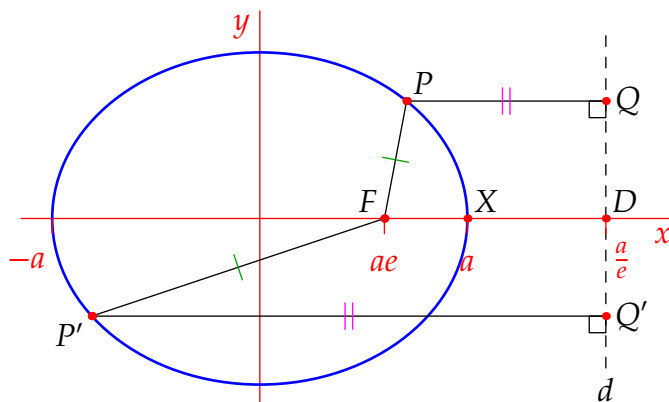
- $e < 1$ yields an *ellipse*; P is *closer* to the focus than the directrix ($|FP|$ is smaller than $|PQ|$).
- $e = 1$ a *parabola*;
- $e > 1$ a *hyperbola*; P is *farther* from the focus than the directrix.

To better understand these, it is helpful to draw axes. We do this explicitly for the ellipse:

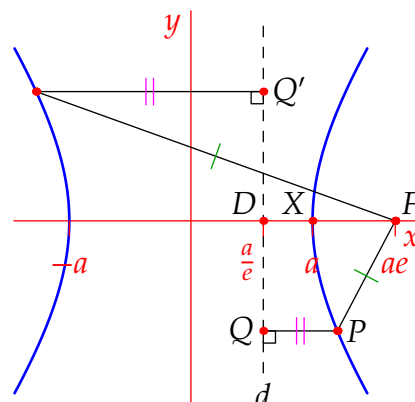
1. Let the x -axis be the perpendicular $\overleftrightarrow{FD}$ to d through F , where D is to the right of F .
2. There exists a point X on the curve *between* F and d , breaking the segment in the ratio $e : 1$.
3. Define $a = \frac{|FX|}{1-e}$ and slide the axis left or right until $X = (a, 0)$.
4. Then $F = (a - |FX|, 0) = (a - a(1 - e), 0) = (ae, 0)$ and the directrix is $x = a + \frac{1}{e}|FX| = \frac{a}{e}$.

The picture is drawn below. In the limit $e \rightarrow 0^+$, the focus becomes $F = (0, 0)$, the point Q moves to ∞ , and the ellipse becomes a *circle* of radius a (circles do not have a directrix).

For the hyperbola, convention places D to the *left* of F (step 1) and defines $a = \frac{|FX|}{e-1}$ (step 3). This results in identical expressions for the focus and directrix.



Ellipse: $e < 1$



Hyperbola: $e > 1$

²³The prefix *para* means *next to*, or *akin to*, alike but not equal: e.g., *parallel*, *paranormal* or *paralegal*. To speak *hyperbolically* is to say more than you mean, to exaggerate. To be *elliptical* is to be brief or condensed, possibly evasive; *ellipses* are dots (...) indicating something omitted.

We need now only re-run the algebra prior to Theorem 6.3 to obtain the familiar co-ordinate form: supposing $e \neq 1$,

$$\begin{aligned}
 |FP| &= e |PQ| \iff |FP|^2 = e^2 |PQ|^2 \\
 &\iff (x - ae)^2 + y^2 = e^2 \left(\frac{a}{e} - x \right)^2 \\
 &\iff (1 - e^2)x^2 + y^2 = a^2(1 - e^2) \\
 &\iff x^2 + \frac{y^2}{a^2(1 - e^2)} = 1
 \end{aligned}$$

Theorem 6.6 (Canonical form of Ellipse and Hyperbola).

Given an ellipse with eccentricity e , there exist rectangular co-ordinates/axes with respect to which the ellipse has equation

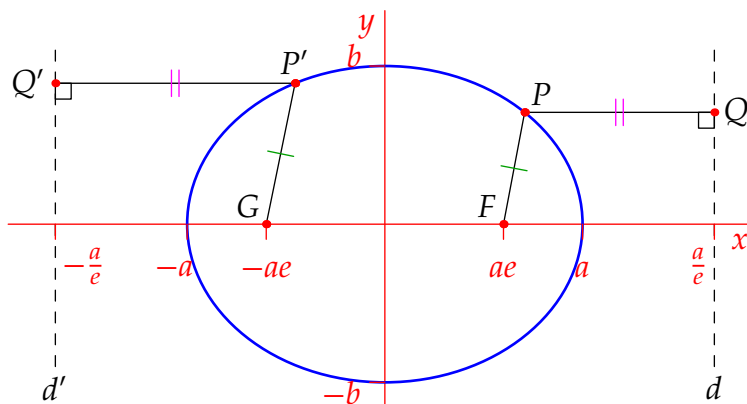
$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1, \quad 0 < b \leq a, \quad e = \sqrt{1 - \frac{b^2}{a^2}} < 1$$

For a hyperbola, the corresponding equation is

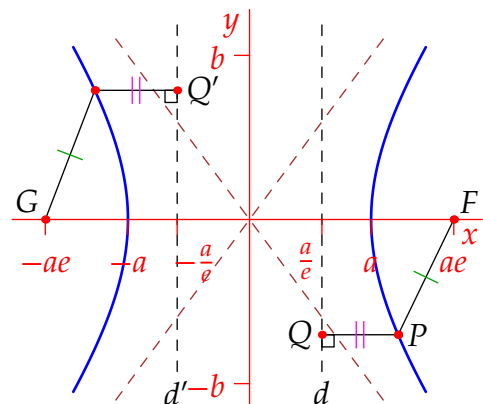
$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1, \quad 0 < a, b, \quad e = \sqrt{1 + \frac{b^2}{a^2}} > 1 \quad (b \text{ can be larger than } a \text{ for a hyperbola})$$

By symmetry in x , both conics have two foci $F, G = (\pm ae, 0)$ and two directrices $x = \pm \frac{a}{e}$. The conics may be defined using either pair.

The values a, b are, respectively, the **semi-major** and **semi-minor** axes.



Ellipse $e < 1$



Hyperbola $e > 1$

The canonical forms make clear why there are precisely two points of an ellipse or hyperbola on the x -axis: $x = \pm a$. For a hyperbola, b is mostly irrelevant except inasmuch as the curve is seen to approach its *asymptotes* $y = \pm \frac{b}{a}$.

Examples 6.7. In general, to put a conic in canonical form we must complete the square *twice*.

1. The canonical form hyperbola

$$\frac{(x+1)^2}{9} - \frac{(y-7)^2}{16} = 1$$

has canonical co-ordinates $X = x + 1$, $Y = y - 7$ and parameters

$$a = 3, \quad b = 4, \quad e = \sqrt{1 + \frac{b^2}{a^2}} = \frac{5}{3}$$

Its foci and directrices are

$$F, G = (-1, 7) \pm (ae, 0) = (4, 7), (-6, 7)$$

$$x = -1 \pm \frac{a}{e} = -1 \pm \frac{9}{5} = -\frac{14}{5}, \frac{4}{5}$$

2. Given the conic with equation

$$9x^2 + 4y^2 - 36x - 8y + 4 = 0 \quad (*)$$

we complete the square.

$$\begin{aligned} (*) &\Leftrightarrow 9(x-2)^2 + 4(y-1)^2 + 4 - 36 - 4 = 0 \\ &\Leftrightarrow \frac{(y-1)^2}{9} + \frac{(x-2)^2}{4} = 1 \end{aligned}$$

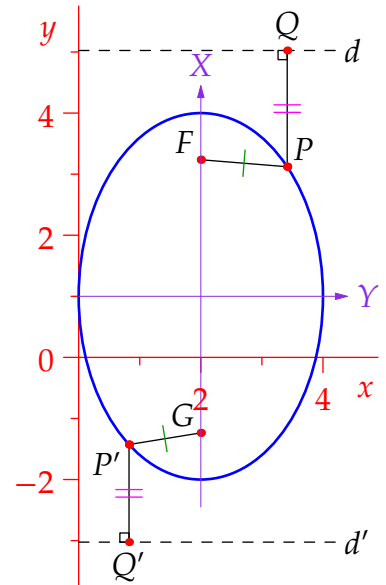
Be careful with the denominators! This is an *ellipse* with canonical co-ordinates $X = y - 1$, $Y = x - 2$ and parameters

$$a = 3 > b = 2, \quad e = \sqrt{1 - \frac{4}{9}} = \frac{\sqrt{5}}{3}$$

The foci and directrices are

$$F, G = (1, 2 \pm ae) = (1, 2 \pm \sqrt{5})$$

$$y = 1 \pm \frac{a}{e} = 1 \pm \frac{9}{\sqrt{5}}$$



Rotated co-ordinates (X, Y) can be used if there is an xy -term, though such is beyond this course. We are now in a position to describe any conic whose equation has the form

$$\lambda x^2 + \kappa y^2 + \sigma x + \tau y + \mu = 0$$

The type of conic depends on the signs of λ and κ .

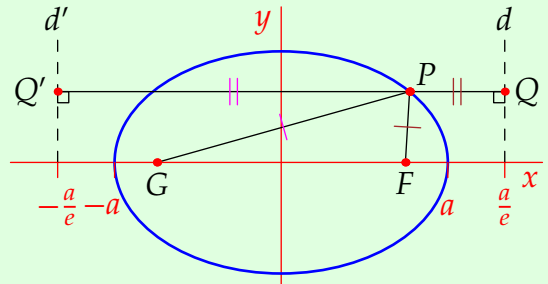
The Reflection Principle for an Ellipse

An ellipse has *two* foci: so what are they focusing? First we make a simple observation.

Lemma 6.8 (Distance Sum from the Foci).

Suppose an ellipse has foci F, G and semi-major axis a . Then the points on the ellipse are precisely those satisfying

$$|FP| + |GP| = e|QQ'| = 2a$$



Imagine holding onto a rope (length $2a$) whose ends are tied to two posts (F, G). The ellipse marks the locations where the rope would be fully taut.

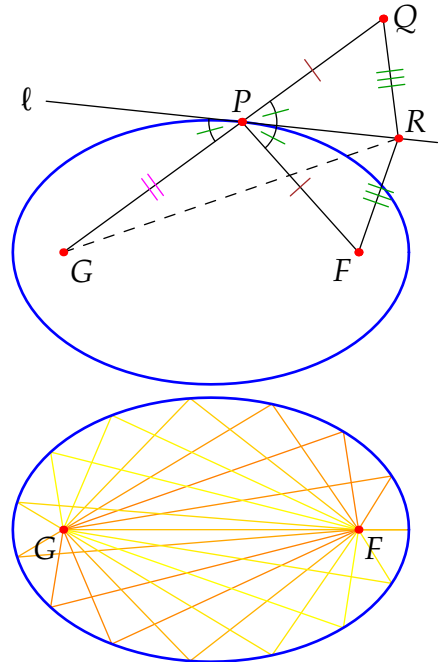
We are now in a position to mimic the parabolic discussion on page 88.

- Given P on an ellipse with foci F, G , extend $\overline{GP}$ to Q such that $|PQ| = |PF|$, and bisect $\angle FPQ$ by a line ℓ .
- If R is any other point on ℓ , then

$$\begin{aligned} |FR| + |GR| &= |QR| + |GR| \\ &\stackrel{\Delta}{\geq} |GQ| = |PQ| + |GP| \\ &= |FP| + |GP| = 2a \end{aligned}$$

where we used the *triangle-inequality*.

- All points $R \neq P$ on ℓ therefore lie *outside* the ellipse, whence ℓ is *tangent* to the ellipse at P .
- If the ellipse is thought of as a mirror, a light source placed at one focus will have its rays reflected onto the other focus.



Equivalently, suppose you were playing mini-golf with the ball at one focus and the hole at the other. You could play your stroke *in any direction* and, provided you hit it hard enough,²⁴ the ball would land in the hole!

Hyperbolæ also have a reflection principle, though it is a little harder to visualise: see Exercise 6.

²⁴And the ball bounced off the wall perfectly so that the angle of incidence equalled that of reflection; due to the complex nature of collisions, this typically *does not happen* in real life.

Exercises 6.2. Key concepts: Eccentricity & definitions of ellipse and hyperbola, Foci and Directrices, Distance sum for an ellipse

- Find the foci and directrices of the hyperbola $\frac{y^2}{16} - \frac{x^2}{9} = 1$. Sketch the curve.
- For each of the following, complete the square to find the foci and directrices. Then sketch the curve.

(a) $x^2 + 4y^2 - 2x + 16y = 17$

(b) $2x^2 - 3y^2 + 8x - 18y = 5$

- The equation $xy = 1$ is a hyperbola. By making the substitution

$$x = \frac{1}{\sqrt{2}}(u - v), \quad y = \frac{1}{\sqrt{2}}(u + v)$$

show that this is indeed the case. Find the foci and directrices of this hyperbola.

(This change of co-ordinates amounts to rotating the plane by 45°)

- As $x \rightarrow \infty$ show that a canonical hyperbola approaches its asymptotes $y = \pm \frac{b}{a}x$.
- Show that the hyperbola with foci F, G and semi-major axis a is the set of points P for which the difference is constant:

$$|FP| - |GP| = \pm 2a$$

- (Hard) Verify the 'reflection' principle for hyperbolæ: if a light source is placed at a focus, then the reflected rays will appear to have emanated from the other focus.
- Let $p \geq 0$ be a constant and consider the intersection of the cone $z^2 = x^2 + y^2$ with the plane $pz = y - 1$.

(a) Describe how the type of conic you obtain relates to the value p .

(b) (Hard) What is the eccentricity of such a conic, and what happens when $p = 0$?

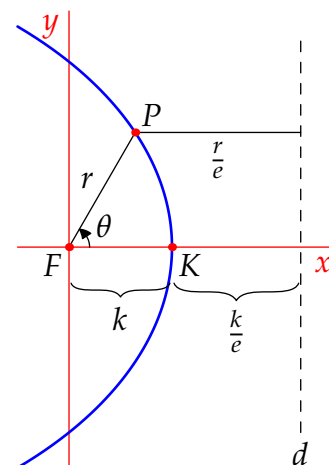
- Place the focus F of a conic with eccentricity e at the origin and orient so that the nearest point of the conic lies on the positive x -axis at $K = (k, 0)$.

(a) Prove that the conic has polar equation

$$r = \frac{k(1 + e)}{1 + e \cos \theta}$$

Describe how the resulting curve depends on e . In particular, explain why don't we get all of a hyperbola when $e > 1$.

(b) Find the rectangular (x, y) equation of this conic.



9. (Hard—if you like Physics) At a point $P = (x, y) = (r \cos \theta, r \sin \theta)$ on a curve (x, y, r, t) are functions of time t , define the unit vectors

$$\hat{\mathbf{r}} = \begin{pmatrix} \cos \theta \\ \sin \theta \end{pmatrix}, \quad \hat{\boldsymbol{\theta}} = \begin{pmatrix} -\sin \theta \\ \cos \theta \end{pmatrix}$$

The first points away from the origin; the second is obtained by rotating the first counter-clockwise 90° .

- (a) Use the chain rule to verify that $\frac{d}{dt}\hat{\mathbf{r}} = \dot{\theta}\hat{\boldsymbol{\theta}}$, where $\dot{r} = \frac{dr}{dt}$ means differentiation with respect to time. What is $\frac{d}{dt}\hat{\boldsymbol{\theta}}$ in terms of $\hat{\mathbf{r}}$?
 (b) Verify that

$$\frac{d^2}{dt^2}(r\hat{\mathbf{r}}) = (\ddot{r} - r\dot{\theta}^2)\hat{\mathbf{r}} + \frac{1}{r} \left[\frac{d}{dt}(r^2\dot{\theta}) \right] \hat{\boldsymbol{\theta}}$$

Note that $r\hat{\mathbf{r}}$ is the position vector of the point P .

- (c) Suppose an object travels along *any conic* whose focus is at the origin in such a fashion that its *speed* $r\dot{\theta} \propto \frac{1}{r}$ is inversely proportional to its distance from the origin: equivalently $r^2\dot{\theta} = L$ is constant.

Use the polar form (Exercise 8) to show that the curve satisfies the differential equation

$$\frac{d^2}{dt^2}(r\hat{\mathbf{r}}) = -\frac{c}{r^2}\hat{\mathbf{r}} \quad (*)$$

where c is a positive constant. Compute c .

This is Newton's model of gravity: following his second law ($\mathbf{F} = m\mathbf{a}$), the object is subject to an *inverse-square* attractive force. If you want a challenge, the converse is also true: $(*)$ implies that the orbit is a conic whose focus is at the origin.

This discussion relates to Kepler's 1st law, that planets orbit the sun in ellipses with the sun at one focus. The constancy of $r^2\dot{\theta}$ is essentially the conservation of angular momentum. The other types of orbit are also possible: while recurring comets (e.g., Halley's) follow elliptic orbits, some have been known to pass the sun along hyperbolic orbits, never to return to the solar system.